Heterogeneous Brownian Motion in Multiple Dimensions

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Abstract

We analyze a model of Brownian motion with several discrete states, each with their own parameters, to calculate its stationary probability and see what states it is likely to be in. Previous research on the topic has been focused upon Lipschitz and continuous states, as opposed to discrete states. We define the model, basing it upon the limit definition, show a case where the stationary probability is constant over each of the discrete states, then show a case where it is not constant, showing it is not as simple as having a state-wide probability. We also run a computer approximation, using the limit definition and Markov chains in order to find solutions of several different cases. We find and show several interesting cases using it.

1 Introduction

Brownian motion is an important phenomenon, that appears in many different fields of science and finance. A natural question that arises when studying probabilistic complex phenomena, based upon Brownian motion, is "What are the steady-state probabilities of events in the given system?" In particular, we are often interested in systems with several, distinct modes, each with its own behavior. For example, we can think of a thermodynamic system of particles where temperature induces different states, or stock prices, where we can think of the market in several different states, each causing specific stocks to have less or more volatility, e.g., the stocks of export-oriented industries may depend on the volatility of foreign affairs. The technique of modeling systems using probability is well-established (e.g., Øksendal's book [Øks00] discusses the case of continuous states). We would like to study the theory of such systems, to know at what state they are likely to be, in order to predict them. In this work, we analyze several cases of Brownian motion systems, and develop algorithms to evaluate their steady state probabilities. Specifically, we consider driftless diffusion over a torus, with variable multidimensional "temperature".

There are several more examples of systems with multiple distinct states. We consider a system inspired by the one outlined in Thapa *et al.*'s work [TPS24]: robots whose locations are determined by Brownian motion, i.e., the speed of each robot depends on whether that robot is close or far from other robots. Each robot has two states: one where it is close to another robot, and one where it is far from all other robots. We would like to find, for example, what is the steady-state probability that many robots are close to each other. However, as we show, the probability density function of the stationary distribution, is not necessarily constant over states, and calculating that probability may

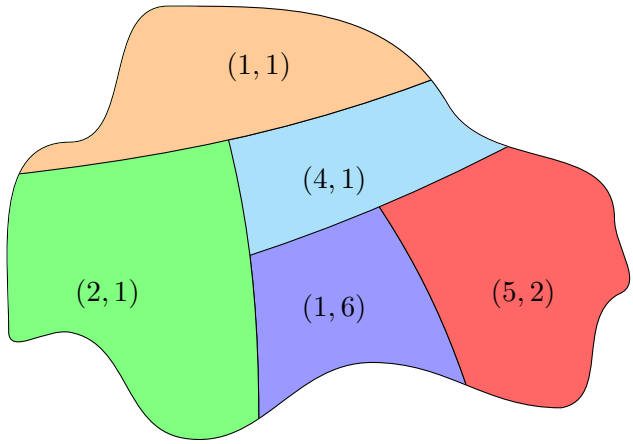

Figure 1: Example of regions. The vector in each region is the BPV.

be nontrivial.

We study an abstract model, henceforth referred to as **Heterogeneous Brownian Motion** where we have a high-dimension torus and a single particle. The space is partitioned into regions, where each region is also a state of the particle in the sense that in each region the particle moves with a different "speed vector" (the Brownian motion parameter) in each dimension. See illustration of an example in Figure 1. We do not consider drift in our model, only diffusion speeds. We henceforth refer to the parameter vector as BPV - "Brownian Parameter vector".

We want to answer questions about the steady state of various cases of Brownian motion - What affects the distribution, whatever it is the shape of the regions, the BPV of the regions, and if it satisfies certain cases - For example, if its constant over regions, and if there is an easy way to calculate and show the steady state without resulting to long random walk simulations.

Note that when considering temperature, more energy means that the particles move faster in all directions. In our model, the particle may move faster only in a specific direction, or move slower in one direction while moving faster in another, etc. Therefore, in our model the BPVs of different regions may be linearly independent, unlike the physical temperature thermodynamic model. In mathematical terms, temperature-like systems have a BPV function that grows isotropically - meaning, it is of the form $v(a) := c(a) \cdot v$ where v is a constant vector, and c is a variable scalar. We henceforth call those systems **linear**, and we refer to system without such BPV functions as **non-linear**. For example, a system comprised of two regions with BPVs $(1, 2)$ and $(2, 4)$ is linear, but one with BPVs $(1, 2)$ and $(2, 1)$ is non-linear.

Our results. We show that in the linear case of heterogeneous Brownian motion, the probability density, evaluated at a location of the particle, of the steady state, is constant in regions - Meaning, for every $x, y \in U$ where U is a region, and for f being the density function of the steady state, that $f(x) = f(y)$. It can also be easily calculated. However, in many cases, even some simple 2D cases with convex regions, this is not true. Specifically, we show that stationary probability density of some specific 2-dimensional systems under this model is not constant per region, using a very simple

counterexample with two regions, with an additional, strictly convex but more complex counterexample in the appendix. In addition, we show a approximation based upon a Markov chain whose steady-state probabilities can be numerically computed. The idea is to represent the "real" torus as a discrete grid of points, each with variable probability to move to neighboring points. The Markov chain is faster to calculate compared to a simulation of the random walk, for certain simple cases, and is deterministic. However, for certain inputs this can result in quantization-related errors.

Paper organization. The remainder of this paper is organized as follows. In Section 3 we introduce the model of heterogeneous Brownian motion and prove some preliminary results. In Section 4 we show that in the linear case, the stationary density is constant in regions. In contrast, we show in Section 5 a simple counterexample in two dimensions, where the density is not constant in regions. In Section 6 we introduce our approximate Markov model and bound its error, In Section 7 we show a few cases to see weaknesses and strengths of that model compared to a standard random walk simulation. Finally, in Appendix 9, we show another, convex counterexample in two dimensions where the desnity of the stationary distribution is not constant in regions, with a more complicated proof. We start with some related work in Section 2.

2 Related work

Brownian motion is an extensively studied area in mathematics and physics. In this section we do not attempt to review it. Rather, we briefly mention results relevant to our work. There are multiple ways to define Brownian Motion, each with its own strengths and weaknesses. We first show them all, then provide detailed explanations.

The gaussian style approach defines Brownian motion as a random process which satisfies a few axioms, namely the Gaussian distribution of differences. This approach is given and elaborated on in the book of Moerters and Peres [MP10]. The Wiener process style approach defines Brownian motion as a limit of a normalized sum of random walks. This style lends itself more easily to defining new processes, as we end up defining heterogeneous Brownian motion with it. This is also well explained in the book of Moerters and Peres [MP10]. A bound on the rate of convergence of random walks to Brownian motion is given in Fraser's work [Fra73], helping us give bounds to the accuracy of our Markov chain based algorithm. Finally, there is an approach of using stochastic differential equations (SDEs): While not directly a component of Brownian motion, SDEs are a basis for many stochastic processes around. Øksendal has written about that formulation in [Øks00].

To generalize Brownian motion to Heterogeneous Brownian Motion, we can look at it from the lens of a general random process. Löcherbach [Loe15] gives a few important results, including the Doeblin condition, Harris recurrence, and convergence in probability. Notably, we use Doeblin condition in order to show ergodicity of Heterogenous Brownian motion, an important result that shows there is a single stationary state.

Definitions based on SDEs tend to look neater, hiding some of the details such as limits by abstracting them into an integral. Øksendal gives in [Øks00] a thorough explanation of that formulation. An important mention in the book is the existence and uniqueness theorem of SDEs for Lipschitz-

Continuous functions - The function presented in our paper, if adapted to an SDE, is not Lipschitz-Continuous, and therefore this theorem does not apply, a reason why we did not define heterogeneous Brownian motion with SDEs.

If we want to define heterogeneous Brownian motion as an SDE, however, we need to have existence and uniqueness of solutions to that SDE, while the SDE may not necessarily have a single, unique solution. In the single-dimensional case, Engelbert proves in [ES85] that driftless SDEs have a unique, single process as a solution, as long as the parameter is above 0 everywhere. Even in cases where it is zero, Engelbert showed exactly what are the possibilities and requirements. However, all of these results are specifically about the $1D$ case. Since we want to study high dimensions too, this result is not enough. We have used a limit based definition instead.

The original motivation for this paper is the work of Thapa et al. [TPS24]. This work introduces a model of robots moving as active Brownian motion, with the speed of a robot determined by its proximity to other robots. We note that we have analyzed non-active Brownian motion in this paper. A major question that appears in the paper is the expected number of robots that are close (by the activation radius of the robot) to each other in a stable configuration. Our goal was to develop tools for analyzing heterogeneous Brownian motion to resolve this question. Each robot has two states, indicating whether it is close to other robots or not. The Cartesian product of the states, then, has 2^n states, where n is the number of robots. Therefore, one can model this system as a single particle, where the location of each robot is represented by the corresponding coordinates of the high-dimensional particle. This single particle then moves as per our model, and thus we can use the tools presented in our paper, such as the Markov chain based simulation, to analyze it.

We have chosen to only deal with relatively simple regions, satisfying some mild topological conditions. This is due to the existence of "traps," shown in Burdzy's work [BCM06]. Burdzy considered reflected Brownian motion, which is a version of Brownian motion where the particle is reflected after hitting a border. The paper shows a geometric construction, where reflected Brownian motion may "get stuck" in a "trap" and never leave a speicfic area, and thus not be recurrent, which would mean it could have multiple, stable configurations, with some states being unreachable from one or the other. Specifically, Burdzy gives an example of a modified Koch starflake, in which the motion may get stuck on the border line. To avoid these cases, we must be careful in the definition of heterogeneous Brownian motion and regions.

3 Model and Preliminary Results

In this section, we first present some notations and definitions for our model of Brownian motion, then present several important traits of it.

3.1 Notation

- The standard measure of $\mathbb{R}^d$ is denoted by μ, i.e., for all measurable $S \subseteq \mathbb{R}^d$, $\mu(S) = \int_{x \in S} 1 dx$.
- For $i \in \mathbb{N}$, we use $[i]$ to denote the set $\{1, 2, \ldots, i\}$.

- For a set U, we denote the indicator function over the set U by $1_U(x) = \begin{cases} 1, & \text{if } x \in U \\ 0, & \text{otherwise} \end{cases}$

- For a vector $s \in \mathcal{S}$ with $\mathcal{S}$ locally isomorphic to $\mathbb{R}^d$, we denote s_j as the projection of s to the jth axis, where $1 \leq j \leq d$.

- We denote the open ball of radius ε centered at x as $B(x, \varepsilon)$. It is defined as $y \in B(x, \varepsilon) \iff |y - x| < \varepsilon$ under the L_2 norm.

- The supremum of a stochastic process X, between times 0 and t is denoted as $\sup(X, t) := \sup\{X(s) \mid 0 \leq s \leq t\}$. The infimum $\inf(X, t)$ is defined similarly.

- The topological closure of a set U is denoted $\overline{U}$, and the topological interior of set U is denoted $int(U)$.

- The **d-dimensional unit torus** is the quotient space of $[0, 1]^d$, where $x \equiv x - \lfloor x \rfloor$ for all coordinates. It is denoted by $\mathcal{T}^d$. (We omit the superscript where it is clear from the context.) We note that $\mu(\mathcal{T}^d) = 1$, and therefore μ is also a probability measure, namely, it is the probability measure of the uniform distribution.

3.2 Brownian motion

We now present the definition of Brownian motion we use. Our basic definition is based upon the one in Moerters and Peres's book [MP10]. We start with standard Brownian motion, then define multidimensional Brownian motion, and finally, define heterogeneous Brownian motion, which is the model we studied.

Definition 1 *A stochastic process $X : \mathbb{R}_{\geq 0} \to \mathbb{R}$ is called (single dimensional)* ***Brownian motion*** *with* ***parameter*** *$v > 0$ if it satisfies the following conditions:*

1. *X is almost surely continuous, i.e., X is continuous with probability 1.*

2. *X has independent increments, i.e., for all $n \in \mathbb{N}$ and for all $t_1 < t_2 < \ldots < t_n$, the random variables $X(t_n) - X(t_{n-1})$, $X(t_{n-1}) - X(t_{n-2})$, $\ldots, X(t_2) - X(t_1)$ are all independent.*

3. *$X(t) - X(s) \sim \mathcal{N}(0, (t-s) \cdot v^2)$ for all t, s with $t > s \geq 0$, where $\mathcal{N} \sim (m, \sigma^2)$ denotes the normal distribution with mean m and variance σ^2.*

Brownian motion with parameter 1 is called ***standard****.*

Definition 2 *Let $d \in \mathbb{N}$. A stochastic process $X : \mathbb{R}_{\geq 0} \to \mathbb{R}^d$ is called a d-****dimensional Brownian motion*** *with parameter $\bar{v} = (v_1, \ldots, v_d)$ if X is a stochastic process such that for each $1 \leq i \leq d$, the projection of $X(t)$ onto its ith dimension, denoted $X_i(t)$, is a Brownian motion with parameter v_i, independent of all other coordinates of X.*

We note that d-dimensional Brownian motion is a cartesian product of single dimensional Brownian motion. We now define the main object of our work, **Heterogeneous Brownian motion**, as follows.

Heterogeneous Brownian motion is a generalization of non-active Brownian motion, such that each point in the space has its own parameter. To simplify, we divide the space into a finite number of regions, where the BPV is the same for every point in a region.

Definition 3 *Let $\mathcal{S}$ be a metric space, with a probability measure μ, locally isomorphic to $\mathbb{R}^d$ for a given $d \in \mathbb{N}$. Let $\mathcal{P} = \{U_1, U_2, \ldots, U_n\}$ be a set of disjoint* ***regions****, where each region is a path-connected regular open subset of $\mathcal{S}$, such that $\sum_{i=1}^{n} \mu(U_i) = \mu(\mathcal{S}) = 1$. In addition, we require $\bigcup_{i<j\leq n} int(\overline{U_i \cup U_j})$ to be path-connected. Let $v : [n] \to (\mathbb{R}^+)^d$ a function called the* ***BPV function****. For any set $U_i \in \mathcal{S}$ we denote $v(U_i) = v(i)$, and for $a \in U_i$ we denote $v(a) = v(i)$. Similarly, for any $a \in \mathcal{S}$ such that $\forall U_i, a \notin U_i$, then $V(a) = \bar{1}$, where $\bar{1}$ denotes the vector of ones. We define the partial sum*

$$Y(n,t) := \sum_{j=1}^{\lfloor t \rfloor} X(n,j),$$

where $X(n,j)$ is a random variable with mean 0 and variance $\frac{1}{\sqrt{n}} \cdot v(X(0) + Y(n,j-1))$, with $X(0)$ distributed in any way we like. Then, heterogeneous Brownian motion is defined by

$$X(t) := \lim_{n\to\infty} (X(0) + Y(n,nt)).$$

We note that the definition above is similar to a wiener process, as stated in [MP10], but adapted for the case of multiple regions.

In addition, we see that the variance of $X(n,j)$, where $X(n,j)$ is as defined above, is upper bounded by $\frac{1}{\sqrt{n}} \cdot max_{x\in\mathcal{S}} v(x)$, and the mean is 0. Therefore, by the dominated convergence theorem, the limit exists almost surely, and therefore the definition is well-founded.

In this work, we often take the metric space to be the d-dimensional unit torus. An important property of Heterogeneous Brownian motion is linearity.

Definition 4 *A heterogeneous Brownian motion X is called* ***linear*** *if its BPV function can be written as $v(i) = c(i) \cdot v$, where v is some fixed vector and $c : [n] \to \mathbb{R}^+$ assigns a positive constant to each region.*

A non-linear heterogeneous Brownian motion is any heterogeneous Brownian motion that does not satisfy this condition.

We now move on to stationary probability.

Definition 5 *Let $X(t)$ be heterogeneous Brownian motion. The* ***stationary PDF*** *(Probability density function) $f_X^*(x)$ of the process $X(t)$, if it exists, is $\lim_{t\to\infty} f_{X(t)}(x)$, where $f_{X(t)}$ is the PDF of $X(t)$ and $x \in \mathcal{S}$. The* ***stationary probability*** *P_X^* of a stochastic process $X(t)$ and a measurable subset $A \subseteq \mathcal{S}$, is defined as $P_X^*(A) = \int_{x\in A} f_X^*(x)dx$.*

We note that there may be several functions which are stationary, and there may be systems which do not have even a limit PDF. For example, Burdzy [BCM06] has shown that a modified form of the Koch snowflake contains "traps," which are enough to show there may be some sets in the region we may never visit. We therefore need a theorem for the existence and uniqueness of such PDF, under our definition.

Theorem 1 *Let X be a heterogeneous Brownian motion. Then, it has a unique stationary distribution f_X^*. In addition, the distribution satisfies the following property : For a stochastic process X with $X(0) \sim f_X^*$, we have that $\forall t > 0, X(t) \sim f_X^*$.*

We first prove a lemma, that intuitively, says that we can reach every point from every point. This is a similar requirement to an ergodic Markov chain.

Lemma 1 *Let $x_0, y \in \mathcal{S}$, and let $t > 0$. Let $f(x_0, t)$ be the probability density function of heterogeneous Brownian motion $X(t)$, with $X(0) = x_0$. Then $f(x_0, t)(y) > 0$.*

Proof: Denote the region of x_0 by U_i, and the region of y by U_j. If $U_i = U_j$ then the lemma follows from the properties of homogeneous Brownian motion: By path-connectedness of U_i, there exists a path p contained in the interior of U_i with endpoints x and y. For $z \in p$, let $\varepsilon_z = \sup\{\varepsilon \mid B(z, \varepsilon) \subseteq U_i\}$, and define $\alpha := \inf\{\varepsilon_z \mid z \in p\}$. Note that for all z, $\varepsilon_z > 0$ because U_i is open, and hence, by continuity and compactness of the path p, we get that $\alpha > 0$. Now, define a cover D' of p as follows: For every $z \in p$, $B(z, \frac{\alpha}{2})$ is a set in the cover. By definition, D' is a cover. As the path is compact, it has a finite sub-cover, say D. As D is finite, we can write it as $D = D_1, D_2, \ldots, D_k$ where $x \in D_1$ and $y \in D_k$, where $D_i \cap D_{i+1} \neq \emptyset$ for $1 \leq i < k$. For any $w \in D_i$, we note that the probability of $\{X(t) \in D_{i+1} \mid X(0) = w\}$ is positive, as $D_{i+1} \subseteq B(w, \alpha)$, from definition. As the probabilities of going from one ball to the next are all positive, by taking the product of the probabilities we get that $f(x, t)(y) > 0$, and we are done for this case.

Now assume that $U_i \neq U_j$. In this case, we construct a graph $G = (V, E)$. The graph's vertices are the regions of $\mathcal{S}$. $(U_i, U_j) \in E$ iff U_i, U_j share a border, i.e, $(U_i, U_j) \in E$ if $A_{ij} := (int(\overline{U_i \cup U_j}) \setminus (U_i \cup U_j)) \neq \emptyset$. Then, as $C_{ij} := A_{ij} \cup U_i \cup U_j$ is open, we can find a ball B around $a \in A_{ij}$ of radius ε such that $B \subseteq C_{ij}$. We let $z_1 \in U_i \cap B$. Then, by the previous argument, there is a path from x to z_1, with positive probability, that never leaves U_i. By Brownian motion, there is a positive probability we reach $B \cap A_{ij}$ from z_1. Then, finally by our definition of heterogeneous Brownian motion, there is positive probability we reach some point $z_2 \in U_j$ from B. Therefore, for any $(U_i, U_j) \in E$, we can "Pass" through an edge in G.

We now show that G must be connected. Let U_i, U_j be regions. We prove there is a path connecting them in G. Denote the union of regions in the connected component of U_i by $U^{'}$, and denote the union of regions not in the connected component of U_i by U^*. Assume that $U_j \in U^*$ for a contradiction. Let $x \in U_i$ and $y \in U_j$. We define $S = \bigcup_{i<j\leq n} int(\overline{U_i \cup U_j})$, and note that by definition of heterogeneous Brownian motion, it is path connected. In addition, $U_i \subseteq int(\overline{U_i \cup U_j})$, as U_i is regular. Therefore there exists a path $\gamma : [0, 1] \to S$ from x to y. Let $\beta = \inf_t(\gamma(t) \in U^*)$. Since every region U is open, then $\gamma(\beta) \notin U$ from continuity. Therefore, $\gamma(\beta) \in A_{kl}$ for some k, l. We proceed by cases. If $U_k, U_l \in U^{'}$, then as $C_{kl} = A_{kl} \cup U_k \cup U_l$ is open, there is a ball $F \subseteq C_{kl}$ around $a \in A_{ij}$ of radius ε. We note $\emptyset = F \cap U^*$, as every U is open. Therefore from continuity, that cannot be as $\gamma(t)$ is an accumulation point of U^*. The case for $U_k, U_l \in U^*$ is similar. Therefore, w.log, $U_k \in U^{'}, U_l \in U^*$. But then A_{kl} is not empty, therefore there would be an edge (U_k, U_l) in the graph, a contradiction. Therefore, there exists some path in the graph $U_i, U_1, U_2, \ldots, U_j$. The probability from x to y would then just be the product of all the crossing and paths, where from a similar argument to before, we see that they are all of positive density. Therefore, $f(x, t)(y) > 0$, as needed. ∎

For the proof of Theorem 1, we use the existing notion of the Doeblin condition and show heteroge-

neous Brownian motion satisfies the condition. Löcherbach [Loe15] has shown that a random process satisfying the Doeblin condition has a single, unique stationary measure. The form of the Doeblin condition we use is as follows -

Definition 6 *Let X be a random process in the metric space $\mathcal{S}$, $x \in \mathcal{S}$. Assume there exists a probability measure ν over $\mathcal{S}$, and a constant $\alpha > 0$, such that $\forall A \subseteq S$ measurable, $P(X(t) \in A \mid X(0) = x) \geq \alpha\nu(A)$. Then, we say X satisfies the Doeblin condition.*

Doeblin condition has the following useful important implication.

Theorem 2 *[Loe15] Let X be a random process satisfying the Doeblin condition. Then, there exist a unique probability measure f_X^* such that :*

- *For every x, if $X(0) = x$, then for every y, $\lim_{t\to\infty} f_{X(t)}(y) = f_X^*(y)$*
- *If $X(0) \sim f_X^*$, then for all t, $X(t) \sim f_X^*$*

We now show that heterogeneous Brownian motion satisfies the Doeblin condition, and as a consequence, Theorem 1 is proved.

Proof: Let $t > 0$. Let $f(x,t)$ be the probability density function of heterogeneous Brownian motion X, with $X(0) = x$. We define the function $g(y) := \inf_{x\in\mathcal{S}}(f(x,t)(y))$. We show that $g(y) > 0$ by contradiction. Assume that $g(y) = 0$ for some y. Then, by definition of infimum there exists a sequence x_n such that $\lim_{n\to\infty} f(x_n,t)(y) = 0$. Since $\mathcal{S}$ is a compact set, there exists a convergent subsequence x_{n_i} such that $\lim_{i\to\infty} x_{n_i} = x$. Then, since heterogeneous Brownian motion is continuous, $f(x,t)(y) = 0$. From Lemma 1 that is impossible and therefore $g(y) > 0$. Similarly, $\inf_y(g,y) > 0$. Define $\beta = \int_{y\in\mathcal{S}} g(y)dy \geq \mu(S)\inf_y(g(y)) > 0$ and a probability measure $\nu(A) := \frac{1}{\beta}\int_{y\in A} g(y)dy$. Then for every x we have

$$\begin{aligned} P(X(t) \in A \mid X(0) = x) &= \int_{y\in A} f(x,t)(y)dy \\ &\geq \int_{y\in A} g(y)dy \\ &= \beta \cdot \nu(A) \end{aligned}$$

Which is the Doeblin condition, completing our proof. ∎

In addition, for standard Brownian motion over a torus, we have the following lemma. We use it in the following section.

Lemma 2 *Let X be a d-dimensional Brownian motion over $\mathcal{T}^d$. Then $\forall x \in \mathcal{T}^d, f_X^*(x) = 1$.*

Proof: We show that if $X(0) \sim U(\mathcal{T}^d)$, then $X(t) \sim U(\mathcal{T}^d)$ as well. We note that Brownian motion is uniformly ergodic and hence the stationary measure is unique. Let $t > 0$ and $x \in \mathcal{T}^d$. Let the PDF

of $X(t)$ be denoted by $f_{X(t)}$. Let $x \in \mathcal{T}^d$. Then, we get the following:

$$\begin{aligned}
f_{X_t}(x) &= \int_{y \in \mathcal{T}^d} f_{X(0)}(y) f_{X(t)-X(0)}(x-y)dy \\
&= \int_{y \in \mathcal{T}^d} f_{X(t)-X(0)}(x-y)dy && f_{X(0)}(y) = 1 \\
&= \int_{z \in \mathcal{T}^d} f_{X(t)-X(0)}(\bar{0}-z)dz && \text{From change of variables, } z = y - x \\
&= \int_{z \in \mathcal{T}^d} f_{X(0)}(z) f_{X(t)-X(0)}(\bar{0}-z)dy && f_{X(0)}(z) = 1 \\
&= f_{X_t}(0)
\end{aligned}$$

This works for any $x \in \mathcal{T}^d$ and therefore f_{X_t} must be constant for any t, and therefore the uniform distribution is stationary. ∎

Now, the question we have is as follows. What is the stationary PDF of heterogeneous Brownian motion? Assuming our BPV function is linear, we know the answer. We show it in the next section.

4 Linear Heterogeneous Brownian Motion

In this section, we state and prove a theorem which allows us to calculate the stationary probability in cases with a linear BPV function.

Theorem 3 *Let $X(t)$ be a linear heterogeneous Brownian motion with system $(\mathcal{P}, v)$, where $\mathcal{S}$ is a torus of some given dimension d. Then, for any region $U^* \in \mathcal{P}$, the stationary distribution $f^*_X(x)$ over U^* is uniform. Moreover,*

$$P^*_X(U^*) = \frac{\frac{\mu(U^*)}{v^2(U^*)}}{\sum_{U \in \mathcal{P}} \frac{\mu(U)}{v^2(U)}}$$

We note that, in addition, this theorem works for any arbitrary single-dimension case. Naturally all parameters are scalar in single-dimension, and by Gale [Gal87], every single-dimensional manifold is a torus, satisfying the premise of the theorem. We first mention a useful connection between the linear heterogeneous Brownian motion and standard Brownian motion.

Definition 7 (Time Transform function) *Given $T \geq 0$, the **time transform** of a process X is defined by $l_X(T) := \int_0^T v^2(X(t))dt$. We write it as $l(T)$ when X is clear from the context.*

From definition, $l(T)$ is piecewise linear, barring a measure 0 subset, due to v being piecewise constant, and that $\frac{dl(T)}{dT} = v^2(X(T))$. Since it is monotonically increasing, its inverse function $l^{-1}(t)$ is well defined. The inverse function has an important property, which we now state and prove.

Lemma 3 *Define the stochastic process $Y(t)$ by $Y(t) := X(l^{-1}(t))$, where X is an arbitrary heterogeneous Brownian motion process. Then, $Y(t)$ is standard Brownian motion process.*

Proof: Let i be any coordinate, and consider the distribution of $Y(t)_i - Y(s)_i = X(l^{-1}(t))_i - X(l^{-1}(s))_i$. We henceforth omit the subscript i. Let $I = [l^{-1}(s), l^{-1}(t)]$. We assume first that for

$\forall \tau \in I$, $X(\tau) \in U_j$. Let $v(U_j)$, the parameter of U_j be denoted by u. By the definition of X, $Y(t) - Y(s)$ is distributed like $\mathcal{N}(0, |I|\, u^2)$, as X is in U_i. We show $|I|\, u^2 = t - s$. By definition of l and under the assumption, then $\frac{dl}{dt} = u^2$. Therefore, $l(l^{-1}(t)) = l(l^{-1}(s)) + u^2 \cdot |I|$. After re-arranging, $t - s = |I|\, u^2$. Therefore $Y(t) - Y(s) \sim \mathcal{N}(0, t - s)$, as required.

Therefore, by definition, Y is heterogenous Brownian motion with BPV 1 in every region, and hence Y is standard Brownian motion. ∎

Finally, to prove Theorem 3 we state the following result from Azema *et al.*

Theorem 4 *[ADR69] For measurable, positive function f over space $\mathcal{S}$:*

$$\lim_{t\to\infty} \frac{1}{t} \int_0^t f(X) = E[f(x) \mid x \in \mathcal{S}].$$

We now prove Theorem 3 using Lemma 3.

Proof: Let $T > 0$. Define $Y(t)$ as in Lemma 3. Then, by Lemma 3, $Y(t)$ is standard Brownian. Since $Y(t)$ is standard Brownian, its stationary distribution is uniform by Lemma 2.

Let $U^* \in \mathcal{P}$. We find $P_X^*(U^*)$. Then, $\mu(U^*) = \lim_{t\to\infty} \frac{1}{l(T)} \int_0^{l(T)} 1_{U^*}(Y(t))dt$ almost surely, from Azema et.al [ADR69], and the fact $\lim_{t\to\infty} l(t) = \infty$. From here, we do change of variables.

$$\begin{aligned}
\int_0^{l(T)} 1_{U^*}(Y(t))dt &= \int_0^{l(T)} 1_{U^*}(X(l^{-1}(t)))dt && \text{From definition of } Y \\
&= \int_0^T 1_{U^*}(X(t)) \cdot v^2(X(t))dt && \text{Substitution of } l(t) \\
&= v^2(U^*) \int_0^T 1_{U^*}(X(t))dt && v \text{ is constant over } U^*.
\end{aligned}$$

We see that this integral is only positive on values where $X(t) \in U^*$, and that $v^2(X(t))$ is constant over those values, equaling $v^2(U^*)$. Therefore, from the two equations above, we get the following equation:

$$\int_0^T 1_{U^*}(X(t))dt = \frac{l(T)\mu(U^*)}{v^2(U^*)} \tag{1}$$

By Theorem 4, $P_X^*(U^*)dt = \lim_{t\to\infty} \frac{1}{T} \int_0^T 1_{U^*}(X(t))dt$, and hence we just have to show T as a function of $l(t)$. We write $T = \sum_{U\in\mathcal{P}} \int_0^T 1_U(X(t))ct$. Therefore we get

$$\begin{aligned}
P_X^*(U) &= \frac{1}{T} \int_0^T 1_{U^*}(X(t))dt \\
&= \frac{\int_0^T 1_{U^*}(X(t))dt}{\sum_{U\in\mathcal{P}} \int_0^T 1_U(X(t))dt} && \text{From linearity of integral} \\
&= \frac{\frac{l(T)\mu(U^*)}{v(U^*)^2}}{\sum_{U\in\mathcal{P}} \frac{l(T)\mu(U)}{v(U)^2}} && \text{From Equation (1)} \\
&= \frac{\frac{\mu(U^*)}{v(U^*)^2}}{\sum_{U\in\mathcal{P}} \frac{\mu(U)}{v(U)^2}},
\end{aligned}$$

As required in (3). To show that f_X^* is uniform, we note that the same proof works for any subset $A \subset U$, with the measure changed accordingly. Therefore our stationary distribution is uniform. ∎

While Theorem 3 holds for any single-dimension space, it does not apply to higher dimensions, because the time transform cannot be applied. In fact, in the next section we show a simple counterexample to it.

5 Heterogenous Brownian motion in multidimensional spaces

In this section, we show a simple example where the heterogeneous Brownian motion stationary probability is not uniform over regions. Specifically, we have the following.

Theorem 5 *For $\mathcal{T}^2$ there exists a system $(\mathcal{P}, v)$ with heterogeneous Brownian motion X such that $\exists U_i \in \mathcal{P}$, with $S_1, S_2 \subset U_i$ with positive measure, and $\frac{P_X^*(S_1)}{\mu(S_1)} \neq \frac{P_X^*(S_2)}{\mu(S_2)}$, i.e, f_X^* is not constant over every single region.*

The construction is as follows.

Definition 8 *The **Slope System** G_σ, defined on the torus $\mathcal{T}^2$, is defined by $\mathcal{P} = (\{U_1, U_2\}, \sigma)$ where $U_1 := \{x, y \mid y < x\}$, $U_2 := \{x, y \mid y > x\}$, $v(1) = (1, 1)$ and $v(2) = (1, \sigma)$.*

We omit the subscript of G_σ where it is clear from the context. The slope system is illustrated in Figure 2.

We show, by contradiction, that Theorem 5 applies in the slope system, with motion $X(t)$. Assume for contradiction that the stationary distribution in the slope system is uniform over each region. Then it must be the case that the stationary distribution is uniform over $\mathcal{T}^2$, as we prove in the following lemma.

Lemma 4 *If for every region U_i, there exists a constant c_i such that $f_X^*(x)$ on G satisfies $\forall x \in U_i . f_X^*(x) = c_i$, then for every pair c_i, c_j, $c_i = c_j = 1$, i.e, f_X^* is constant over $\mathcal{S}$.*

Proof: Assume the upper and lower regions are uniform, as per the lemma. By integrating over $\mathcal{T}^2$, we get

$$1 = \int_{(x,y)\in\mathcal{T}^2} f_X^*(x, y) = \int_{(x,y)\in U_1} c_1 + \int_{(x,y)\in U_2} c_2 = \frac{c_1 + c_2}{2} \ ,$$

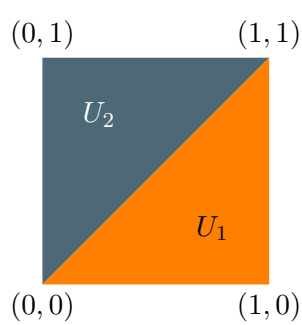

Figure 2: The Slope configuration, the border line is $x = y$.

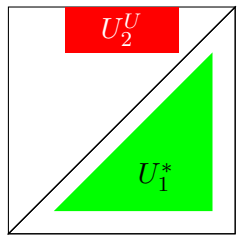

Figure 3: Areas U_2^U and U_1^* we define for the proof.

which implies $c_2 = 2 - c_1$. Denote the PDF of X_1 by g. Note that the stationary distribution of $X(t)_1$ is uniform everywhere. This is because the parameter of the first coordinate always equals 1 and thus, the projection of $X(t)$ to the first coordinate is a standard Brownian motion over $\mathcal{T}^2$, and the stationary distribution for standard Brownian motion is uniform, by Lemma 2. Therefore, $g(x)$ is constant. Moreover, $\forall x \in [0,1]$,

$$\begin{aligned} 1 = g(x) = \int_0^1 f(x,y)dy = \int_0^x c_1 dy + \int_x^1 (2-c_1)dy = c_1 x + (2-c_1)(1-x) \\ = 2 - c_1 - 2x + 2c_1 x \end{aligned}$$

From this, it follows that $1 = 2 - c_1 - 2x + 2c_1 x$, and therefore either $x = \frac{1}{2}$ or $c_1 = 1$. Since the equation needs to be satisfied for any $x \in [0,1]$, that implies $c_1 = 1$. From the formula, $c_2 = 2 - c_1 = 1$. Therefore the stationary PDF of the entire region is constant. ∎

To reach contradiction, we now show that the stationary distribution of the slope system cannot be uniform. We denote the uniform PDF over the Torus as $\text{Unif}(\mathcal{T}^2)$. To this end, we first show a lower bound for $P(X(t) \in U_1 \mid X(0) \sim \text{Unif}(\mathcal{T}^2))$, and show that given some specific parameters, it is larger than $\frac{1}{2}$, a contradiction to Lemma 4. We see that $P(X(t) \in U_1) = P(X(0) \in U_1) = \mu(U_1) = \frac{1}{2}$ due to Lemma 4 and that $X(0) \sim \text{Unif}(\mathcal{T}^2)$. We see that if $X(0)$ is at a point sufficiently far from U_2, X has a high chance to stay in U_1 in time t. We define that set of points as follows (See Figure 3)

$$U_1^*(c,t) := \left\{x > y - 4c\sqrt{t}\right\} \cap \left\{y > 2c\sqrt{t}\right\} \cap \left\{x < 1 - c\sqrt{t}\right\} \ ,$$

where c is some arbitrary positive constant and t is our time parameter.

The probability of entering if $X(0) \in U_1^*$ is bounded by the following lemma.

Lemma 5 $\Pr(X(t) \in U_1 \mid X(0) \in U_1^*) > 1 - 4\Phi(-c)$.

Proof: X is almost surely continuous, and hence if $X(t) \in U_2$ and $X(0) \in U_1$, there must be some $s < t$ such that either $X(s) \in \{(x,y) \mid x = y\}$, or $X(s)_1 = 1$, or $X(s)_2 = 0$, since U_2 is bounded by these 3 lines. We show that the probability of reaching one of them is low (the others are proven similarly). Assume $X(0)_1 < 1 - c\sqrt{t}$. Let Y be a standard Brownian motion, and note that assuming $\forall i X(i) \notin U_2$, then $X \sim Y$. We note according to Y, and according to the reflection lemma, $\sup(Y_1, t) \sim 2\mathcal{N}(X(0)_1, t)$. Therefore from the normal CDF (Cumulative distribution function) we get that $\Pr(\sup(Y_1,t) \geq 1) = 2\Phi(-c)$, as required. The proof is similar for the other two borders, with the total coming from the Union bound. ∎

Lemma 5 gives us a lower bound on U_1^* when it comes to staying inside U_1. This is one of the two lower bounds we need. For the second, define

$$U_2^U = \left\{0.25 \le x \le 0.75, y > 1 - d\sigma\sqrt{t}\right\} \ ,$$

where d is some arbitrary positive constant, σ is the parameter from G_σ, and t is the time parameter (see Figure 3).

Lemma 6 $\Pr(X(t) \in U_1 \mid X(0) \in U_2^U) > \left(1 - 2\Phi(\frac{0.05}{-t})\right)\frac{2\Phi(-d)}{3}$, *where* Φ *is the normal cdf.*

Proof: First, we only consider the cases where there is no s such that $X(s)_1 \notin [0.2, 0.8]$. As X_1 is standard Brownian motion, according to the reflection principle, $\sup(X_1, t) \sim 2\mathcal{N}(X(0)_1, t)$. Therefore $\Pr(\sup(X_1, t) > 0.8) < \Phi(\frac{0.05}{-t})$. Similarly for the infinum. We again separate the probability into two parts.

$$\Pr(X(t) \in U_1 \mid X(0) \in U_2^U) \ \ge \ \Pr(X(t) \in U_1 \mid \exists s \le t, X(s)_1 = 0) \cdot \Pr(\exists s \le t, X(s)_1 = 0 \mid X(0) \in U_2^U) \ .$$

The two factors are bounded separately. For $\Pr(\exists s \le t, X(s)_1 = 0 \mid X(0) \in U_2^U)$, define the process Y such that it is a homogeneous Brownian motion with parameters $(1, \sigma)$. Note that $Y \sim X$ as long as $X \in U_2$, therefore $\sup(Y_2, t) \sim \mathcal{N}(X(0)_2, \sigma\sqrt{t})$, therefore with probability at least $2\Phi(-d)$ we have $\sup(Y_2, t) > 1$ which means there is some s such that $X(s)_2 = 1$. From the fact $X_1 \in [0.2, 0.8]$, then $X \in U_2$.

For the other probability, $\Pr(X(t) \in U_1 \mid \exists s \le t, X(s)_1 = 0)$, we note again from the border definition when $t \to 0$, $\Pr(X(t) \in U_1 \mid s \le t, X(0)_1 = 0) \to \frac{1}{2}$ (by symmetry), so there is some t such that $\Pr(X(t) \in U_1 \mid s \le t, X(0)_1 = 0) > \frac{1}{3}$. By conditioning on the integral

$$\begin{aligned}
&\Pr(X(t) \in U_1 \mid \exists s \le t, X(s)_1 = 0) \\
&\qquad = \int_0^t \Pr(X(t) \in U_1 \mid X(a)_1 = 0) \ \cdot \ \Pr(X(a)_1 = 0 \mid \exists s \le t, X(s)_1 = 0) da \\
&\qquad = \int_0^t \Pr(X(t-a) \in U_1 \mid X(0)_1 = 0) \ \cdot \ \Pr(X(a)_1 = 0 \mid \exists s \le t, X(s)_1 = 0) da \\
&\qquad > \int_0^t \frac{\Pr(X(s)_1 = 0 \mid \exists s \le t, X(s)_1 = 0)}{3} ds \\
&\qquad = \frac{1}{3} \ ,
\end{aligned}$$

we get the result. ∎

The bounds proved in the above two lemmas contradict Lemma 4, because

$$\begin{aligned}
\Pr(X(t) \in U_1) > \ & \Pr(X(t) \in U_1 \mid X(0) \in U_2^U)\mu(U_2^U) + \Pr(X(t) \in U_2 \mid X(0) \in U_1^*)\mu(U_1^*) \\
> \ & \Pr(X(t) \in U_1 \mid X(0) \in U_2^U)\mu(U_2^U) + (1 - 8\Phi(-c))\mu(U_1^*) \\
> \ & (1 - 2\Phi(\frac{0.05}{-t}))\frac{2\Phi(-d)}{3} \cdot \frac{1}{2} d\sigma\sqrt{t} \ + \ (1 - 8\Phi(-c))\mu(U_1^*) \\
= \ & (1 - 2\Phi(\frac{0.05}{-t}))\frac{1\Phi(-d)}{3} \cdot d\sigma\sqrt{t} \ + \ (1 - 8\Phi(-c))\frac{(1 - 7c\sqrt{t})^2}{2} > \frac{1}{2} \ .
\end{aligned}$$

For small enough t, large enough σ, large enough c and small enough d we get the contradiction. For example, $t = 10^{-12}$, $\sigma = 2000, c = 10, d = 1$. Therefore, from the definition of the stationary probability, the constant function cannot be stationary, and the proof of Theorem 5 is complete. ∎

However, even when we cannot calculate the probabilities, we would like some way to approximate the stationary probability. We propose a method based upon finite size Markov chains.

6 Simulation

In this section we consider the question of approximating the steady state probabilities of heterogeneous Brownian motion. The straightforward method is to run a simulation of a particle that approximates the process, but this method requires very long runs due to the large mixing time, inversely proportional to the variance (Which itself is inversely proportional to the noise) of the simulation. Instead, we propose to approximate the system as a discrete Markov chain and find its steady state probabilities algebraically. Since this involves solving high-degree polynomials, the solution is obtained numerically and results in no noise. We first consider, in Section 6.1, homogeneous Brownian motion, and then extend it, in Section 6.2, to heterogeneous Brownian motion model.

6.1 Markov chains for homogeneous Brownian motion

We are interested in simulating Brownian notion.

We use the definition of Brownian motion as the limit of random walks. Consider the random process $X_k(t)$, defined as follows, for $k \in \mathbb{N}$.

$$X_{k^2}(t) := \frac{1}{k} \sum_{i=1}^{k^2 t} X_i \ , \tag{2}$$

where the X_is are i.i.d. variables with variance 1 and mean 0. Donsker's Theorem (see, e.g., Moerters [MP10]) states that $X_{k^2}(t)$ converges in distribution to a standard Brownian process when $k \to \infty$. Clearly, if we multiply the sum in the r.h.s. of Eq. (2) by v for some $v > 0$, and let $k \to \infty$, the sum converges to a Brownian process with BPV v: this can be viewed as a change of (distance) unit.

The idea in the simulation is to quantize the underlying metric space using a grid of K^d points for some $K \in \mathbb{N}$, and replace the almost surely continuous Brownian motion with a discrete process that jumps between points of that grid. Specifically, if the process is at a grid point $(a_1, a_2, \ldots, a_d) \in \{0, \frac{1}{K}, \ldots, \frac{K-1}{K}\}$, each of its coordinates, independently, is incremented by $\pm\frac{1}{K}$, that corresponds to that coordinate, as given by Donsker's theorem.

Formally, we define the Markov chain as follows. Let $K \in \mathbb{N}$. the states of the chain are all points in the set $\mathcal{I} := \left\{ \frac{i}{K} \mid i \in \mathbb{Z}, 0 \leq i < K \right\}^d$. The transition probabilities are as follows. For homogeneous motion, the probability to move in either direction is $\frac{1}{2}$, for a regular Brownian motion.

Definition 9 (Transition Matrix) *The transition matrix of resolution K, denoted by $W[K]$, is*

defined by the following transition probabilities for $a, b \in \mathcal{I}$

$$\Pr(a \to b) = W_{a,b} := \prod_{i=0}^{d} f(a, b, i),$$

where

$$f(a, b, i) := \begin{cases} \frac{1}{2}, & \text{if } |b - a|_i \in \left\{\frac{1}{K}, \frac{K-1}{K}\right\} \\ 0, & \text{otherwise} \end{cases}$$

For a matrix W of size $Q \times Q$, and vectors $a, b \in Q$, we denote $W_{a,b}$ as the entry in W where the vector a corresponds to the entries column, and vector b corresponds to the entries row. We may use 0 here to denote the zero vector. Similarly we denote $(W)^n$ as W to the power of n. We note that this Markov chain is strongly connected, by definition (strongly connected graph with positive weights on each edge). It may, however, be periodic, but we show later that this is not the case. Using Definition 9, we now define a corresponding stochastic process as follows.

Definition 10 *Let* $W[K]$ *be as in Definition 9. The process* $Z[K]$ *is given by* $Z[K](0) = 0$*, and* $\Pr[Z[K](t) = b \mid Z[K](t - \frac{1}{K^2}) = a] = W_{a,b}$*, and* $\forall a < \frac{1}{K^2}$*,* $Z[K](t + a) = Z[K](t)$*.*

We note that by definition, $Z[K](\frac{t}{K^2})$ is distributed like the matrix $W[K]^t$.

Theorem 6 (Matrix-Brownian Limit) *As* $K \to \infty$*,* $Z[K]$ *converges in probability to a Brownian motion.*

We first prove that W^n is distributed identically to $\sum_{i=1}^{n} X_i$, for independent variables X_i, defined as being either $\frac{1}{K}, -\frac{1}{K}$, each with probability $\frac{1}{2}$.

Lemma 7 *For all* $n \in \mathbb{R}$*,* W^n *is distributed like* $\sum_{i=1}^{n} X_i$*, namely,* $\forall a, (W^n)_{0,a} = \Pr(\sum_{i=1}^{n} X_i = a)$*.*

Proof: Let $P(n, z) := \Pr(\sum_{i=1}^{n} X_i = z)$. We prove the lemma by induction on n. The base case is by definition of W. For the inductive step, $W_{0,a}^n = \sum_{k \in \mathcal{S}} W_{0,k}^{n-1} W_{k,a} = \sum_{k \in \mathcal{S}} P(n-1, k) P(1, |k - a|) = P(n, a)$, as required. We note that $W_{k,a} = W_{0,|k-a|}$ from symmetry. ∎

We now prove Theorem 6.

Proof: By Lemma 7, $Z[K](t)$ is distributed like $\sum_{i=1}^{\lfloor K^2 t \rfloor} X_i$. We define $Y_i := K \cdot X_i$ and note that $\sum_{i=1}^{n} X_i = \frac{1}{K} \sum_{i=1}^{n} Y_i$. We note that Y_i is distributed as follows.

$$Y_i := \begin{cases} 1, & \text{with probability } \frac{1}{2} \\ -1, & \text{with probability } \frac{1}{2} \end{cases}$$

Therefore, from Donkser's theorem, as $K \to \infty$, $\frac{1}{K} \sum_{i=1}^{\lfloor K^2 t \rfloor} Y_i \to X(t)$, where X is standard Brownian motion with $X(0) = 0$. Therefore, $\lim_{K \to \infty} Z[K](t) = \lim_{K \to \infty} \sum_{i=1}^{\lfloor K^2 t \rfloor} X_i = X(t)$. Therefore, the stationary distribution of $\lim_{K \to \infty} Z[K](t)$ is the stationary distribution of Brownian motion, and switching limits around we get that the limit of stationary distributions equal the stationary distribution of Brownian motion, as we wanted. ∎

Fraser gives an upper bound on the inaccuracy the approximation used.

Theorem 7 ([Fra73]) *Let $X_k(t)$ be as in Eq. (2) where $\forall i \exists c > 0$ s.t $E[|X_i|^c] < \infty$. Let $X(t)$ be a Brownian motion process as in Definition 1. Then for any $0 \le t \le 1$ and any $q \in \mathbb{R}$,*

$$\Pr\left[|X(t) - X_{K^2}(t)| > \alpha \frac{\log^{\frac{3}{2}+\frac{1}{c}} K}{\sqrt{K}}\right] = O\left(\frac{1}{K^q}\right)$$

where α depends only on c and q.

Note that the bound above converges relatively slowly. In addition, it may generate "artifacts" of which we see in the next section.

6.2 Markov chains for heterogeneous motion

We now extend the results of the previous subsection to heterogeneous Brownian motion.

To approximate the steady state probabilities, we use the following algorithm.

1. We define a sequence of random walks, that converges to a heterogeneous Brownian process with the corresponding parameter for each region.
2. Construct a Markov chain that corresponds to this random walk.
3. Find the eigenvector of the Markov chain, i.e., the stationary probability of the Markov chain.

Each point s is endowed with a BPV $v(s)$ it inherits from the heterogeneous Brownian motion process, according to the region in which s is located (i.e., if $s \in U_i$ then $v(s) = v(U_i)$). For each dimension j, let $v_j(s)$ denote the value of the j coordinate in the BPV $v(s)$. Without loss of generality, let us assume that $v_j(i) < 1, \forall i \le r, \forall j \le d$. (Otherwise, we divide all parameters by twice their maximum). When all parameters are positive, this assumption is equivalent to just changing the distance units. The probability to move in either direction is $\frac{v_j(s)^2}{2}$, and the probability that the i dimension remain unchanged is $1 - v_j(s)^2$. The movement in each dimension is independent of all other dimensions.

To define for the Markov chain, we define the states of the chain $\mathcal{I}$ to be the same as in Section 6.1. We follow by the transition probabilities, as follows.

We note that $X_{a,i}$ has mean 0, and variance $\frac{v^2}{K^2}$. An infinite sum of them would therefore converge according to Donkser's theorem.

Definition 11 *The Heterogeneous transition matrix W of resolution K is defined as follows:*

$$\Pr(a \to b) = W_{a,b} := \prod_{i=0}^{d} f(a,b,i),$$

where

$$f(a,b,i) := \begin{cases} \frac{v_{a_i}^2}{2}, & \text{if } |b-a|_i \in \left\{\frac{1}{K}, \frac{K-1}{K}\right\} \\ 1 - v_{a_i}^2, & \text{if } |b-a|_i = 0 \\ 0, & \text{otherwise} \end{cases}$$

We also define $Z[K]$ identically to the last subsection - Refer to Definition 10. We note that this Markov chain is strongly connected, by definition (Strongly connected graph with non zero edge weights). It is aperiodic, due to having self loops with non zero weight in every node. Therefore it is ergodic.

Theorem 8 (Matrix-Brownian limit) *As $K \to \infty$, $Z[K]$ converges to a heterogeneous Brownian motion.*

To prove the theorem, we use the notion of the random walk variable in a region.

Definition 12 *The random walk variable in region a is defined as follows:*

$$X_{i,j} := \begin{cases} \frac{1}{K}, & \textit{with probability } \frac{v_{i_j}^2}{2} \\ -\frac{1}{K}, & \textit{with probability } \frac{v_{i_j}^2}{2} \\ 0, & \textit{with probability } 1 - v_{i_j}^2 \end{cases}$$

Similarly to the previous subsection — If we find W^n, we get that it has the same distribution of $\sum_{j=1}^{n} X_{i,j}$, assuming we did not cross regions, where $X_{i,j}$ is defined as in Definition 12. The Lemma and proof are identical to Lemma 7. Therefore $Z[K]$ is distributed like $\sum_{i=0}^{\lfloor K^2 t \rfloor} X_i$. By using the lemma we then get our proof.

Proof: We note that $X_{i,j}$ has variance $\frac{v_{i_j}^2}{K^2}$. By the definition of heterogenous Brownian motion, we define $Y(K^2, t) := \sum_{j=1}^{t} X_{i_t}$ where $i_t = Y(K^2, t-1)$. Then, $Y(K^2, K^2 t)$ converges to Heterogenous Brownian motion. This is simply the sum of X_i, similar to the previous subsection. From this and our definition we get that $Z[K]$ converges to Brownian motion, as we require. Therefore the eigenvectors of W^{K^2}, and thus the eigenvectors of W tend to show our stationary distribution over our Heterogeneous Brownian motion, as we require. ∎

Note that there are K^d states in the Markov chain, so its transition probability matrix is of size $K^d \times K^d$. To approximate the stationary distribution of a given heterogeneous Brownian motion, we find the eigenvector of the largest eigenvalue of the transition matrix described above.

We have designed a program in Python in order to build then solve the Markov Chain to find the eigenvalues, and thus, steady state. The program we have designed first builds the matrix, then finds its eigenvectors using the function `numpy.linalg.eig`, which uses the standard LAPACK routines to solve for the eigenvalues [ABB+99]. This only works in practice for very small matrices, up to about $K = 100$ without blowing up system memory, due to the quartic dependency on K. However, even such small parameter is large enough for many functions in practice. On an Intel Core i7-12700 with 16GB of DDR4-3200, running Linux 6.6.87, the run times were as follows:

K	Matrix ($K^2 \times K^2$) generation time (seconds)	Eigenvalue solve time in seconds
10	0.010828327000012905	0.03093619599997055
20	0.011574864999943202	0.06732398100007231
40	0.0378064499999482	1.072088979
80	0.13971452199996293	83.328634577

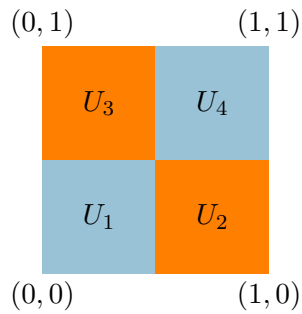

Figure 4: The Grid configuration. Regions with the same color have the same BPVs.

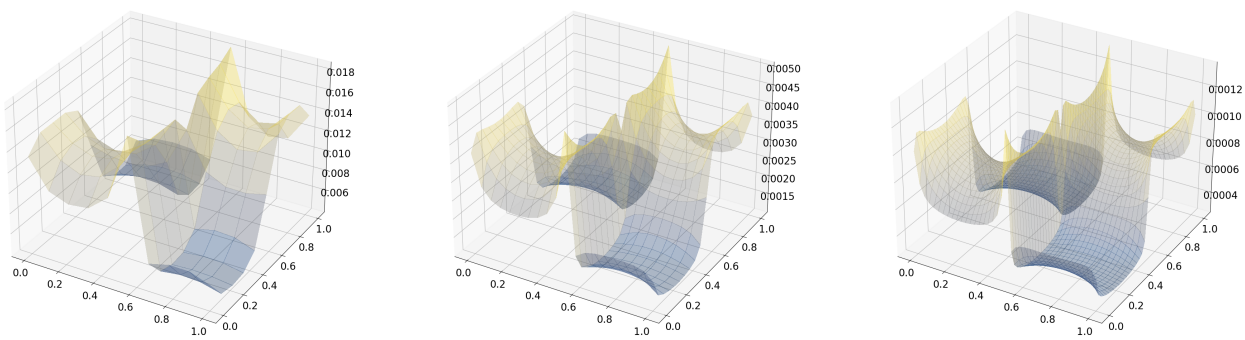

Figure 5: Steady state distribution in the grid configuration with three quantizations: from left to right, $K = 10$, $K = 20$ and $K = 40$. Graph of density as function of location.

7 Case studies

In this section we show several examples of the Markov solver working for simple cases. We analyze different K and see how the graphs change in response. Specifically, we consider the "grid" system, where the torus is partitioned like a 2×2 checkerboard (Figure 4), and the "Coil-n" system, where the torus is partitioned into two connected spirals (Figure 8).

The 2×2 grid. The grid partitions the unit 2-dimensional torus along the lines $x = 0, x = 1/2, y = 0, y = 1/2$ (see Figure 4). We denote the bottom left quadrant by U_1, the bottom right by U_2, top left by U_3 and top right by U_4. In our example, we fix the BPV of U_1 and U_4 as $(1, 1)$ and the BPV in U_2 and U_3 to be $(2, 1)$. Note that the BPV in the second coordinate is always 1. Figure 5 shows the results of our simulations with 3 different quantizations. It is interesting to observe the highly non-constant values of the steady state probabilities as they are reflected in the charts.

Convergence rate and the slope model. It is interesting to compare the empirical results to the bound from Theorem 7. For a bound of 0.1 from the real distribution, we need approximately $K^{-\frac{1}{2}} = 0.1$, or $K = 100$, already the upper bound of what we can calculate for a very rough bound. In practice, the empirical results are better.

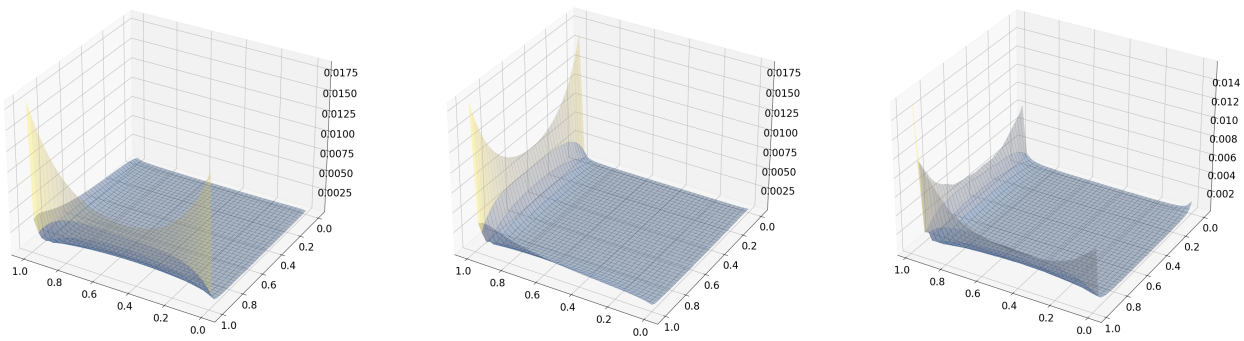

Figure 6: Slope configuration, quantization $K = 40$. BPV on the diagonal is $(8, 1)$ in the left chart, and $(1, 8)$ in middle. Right: simulation result. Graph of density as function of location.

The charts in Figure 5 seems to indicate better convergence. However, there are many other cases for which we seem to get comparatively worse results, possibly suggesting that in certain cases Fraser's bound is tight. For example, consider the "slope" configuration used in Section 5 (Definition 8). Refer to Figure 2). When the BPV is $(8, 1)$ in U_1 and $(1, 8)$ in U_2, we see two different charts depending on the value of the BPV on the diagonal, shown for quantization parameter $K = 40$ in Figure 6. In addition, we have run a simulation of a Brownian motion to get a sense of the ground truth. The simulation used a normal distribution with small variance. The normal distribution was of parameter $\frac{v}{10^6}$, where we have run the simulation for 10^6 such steps, with our initial point being selected uniformly. We have ran it 100 times, then removed 5 outliers to increase readability and calculated the average to get the chart on the right.

Naturally, both the simulation (as the simulation time goes to infinity) and the Markov chain (as the quantization parameter K goes to ∞), yield the same result in the limit. In random walk simulations (right chart in Figure 6, we can see two "walls" at the axes, while the Markov approximation (Figure 6, left and center charts) produces one large "wall" and another small wall, whose location depends on the association of the diagonal entries in the matrix. However, as predicted by Theorem 7, the height of the two walls in the Markov simulation get close as K grows: see Figure 7.

The coil(n) model. We also consider the model we call coil(n), which may resemble a ribbon that goes around the torus n times. Formally, coil(n) is defined as follows: the torus is partitioned in two regions:

- $\{(x, y) \mid (nx - y) - \lfloor nx - y \rfloor < 0.5\}$, and
- $\{(x, y) \mid (nx - y) - \lfloor nx - y \rfloor > 0.5\}$.

The graph is illustrated in Figure 8.

We simulated and solved the Markov approximation for the coil(2) case, where one continuous region has BPV $(8, 1)$ and the other $(8, 8)$. This example is interesting because it results in uniform distribution in each region, as can be seen in Figure 9.

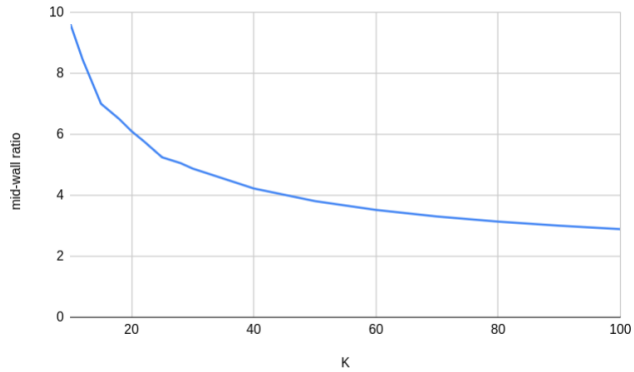

Figure 7: Markov simulation of the slope model: Ratio of the probability of point $[0, \frac{1}{2}]$ to point $[\frac{1}{2}, 0]$ under the Markov solver, as a function of the quantization parameter K.

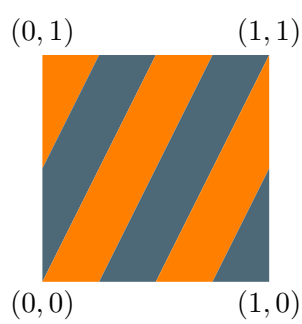

Figure 8: The coil(2) configuration. The BPVs are $(8, 1)$ and $(8, 8)$.

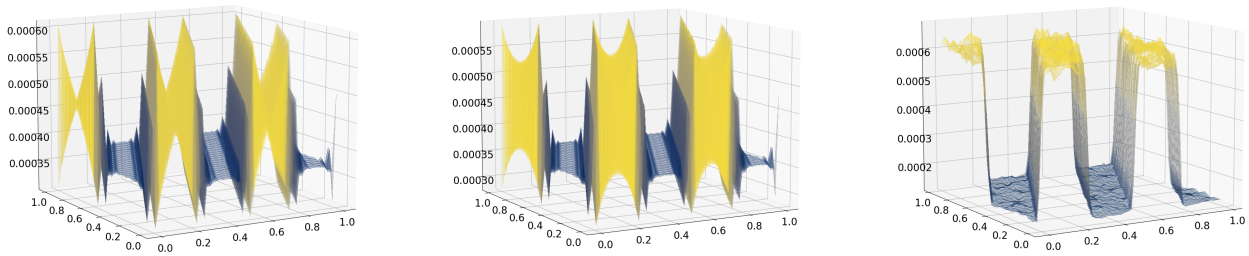

Figure 9: $K = 49$ on left, $K = 50$ on middle and simulation on right. Graph of density as function of location.

We note that when the bright areas have BPV (x_1, y_1) and the dark areas have BPV (x_2, y_2), both the simulation and the Markov chain analysis seem to indicate that the stationary probability of each area $i \in \{1, 2\}$ tends to $\frac{(nx_i)^2+y_i^2}{(nx_1)^2+(nx_2)^2+y_1^2+y_2^2}$,where n is the slope of the region border. This was tested upon multiple slopes and parameters.

The Markovian model has two different problems here. First, it oscillates, and second, there are artifacts (similar effects occur with Fourier transform, e.g., Gibbs phenomenon). The artifacts shrink as $K \to \infty$. The oscillation can be seen in Figure 9.

We note here that the artifacts are very sensitive to the value of the quantization parameter K. We can see the significant difference between results for $K = 49$ and $K = 50$ in Figure 9. We had also run a simulation (with the same simulation parameters used in the slope case), which exhibits significantly less artifacts here, as shown in the right side of the figure. This shows that for cases where the quantization significantly affects the space, a random walk simulation may be preferred.

Finally, let us note that there are cases in which the stationary distribution is uniform even though the parameters are different. For example, the coil configuration with BPVs $(1, 1)$ and $(\frac{1}{2}, 2)$. If we run a random walk simulation, we easily see the stationary distribution is uniform, as in Figure 10. This is consistent with the formula above.

8 Summary

In this paper we defined and studied with Heterogeneous Brownian motion. More specifically, we defined the model, proved it is ergodic under some mild conditions, showed that in simple cases (called "linear") it is easy and simple to calculate its stationary distribution function. We also showed that in some non-linear cases, the stationary distribution is much more complex, with a simple counterexample. In addition, we gave a simple simulation idea and tool for approximating the stationary distribution function.

We may conclude that even simple random process, such as the slope case presented in Section 5 can have wildly unexpected stationary distributions, and, conversely, that cases such as the wrap

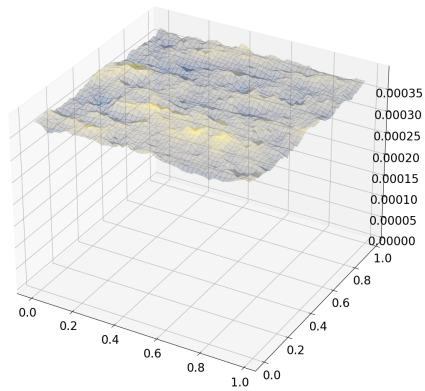

Figure 10: Random walk simulation of coil-2 case with BPVs (1,1) and (0.5,2). Graph of density as function of location.

case from Section 7, have simple stationary distributions. This has wide implications, as it shows the intractability of such problems.

We have only made a first small step in studying the model. Many questions remain open.For example, in Section 7, we saw a case in which the stationary distribution appears flat despite being non linear. We do no have a full analysis of this case. Second, there are no closed form formulas for any of the complicated cases - There may be hiding a formula, for, e.g, the grid case. In addition, the paper assumed there are no non-zero covariances. It may be interesting to look into cases where the covariances are not zero and see if there are hidden simple cases there.

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9 Appendix

In the appendix, we show another example of a case where the stationary probability density function is not constant over regions. This case is convex, but more complicated than the slope case. We start by defining the grid case, and the associated notation $G(v)$.

Definition 13 *We define the grid case of the torus by Figure 4, where $U_1, ...U_4$ are the areas shown - U_1 is bottom left, U_2 is bottom right, U_3 is top left and U_4 is top right area of the torus.*

Definition 14 *We define $G(v)$ to be a heterogeneous Brownian Motion, in the grid case, where areas U_1, U_4 have BPVs $(1, v)$ and areas U_2, U_3 have BPVs $(v, 1)$.*

We show the theorem Theorem 5 for the grid case. The proof is by contradiction. We first observe the following - Lemma 4 works in the grid case.

Proof: By symmetry. A torus centered on $(0, 0)$ - i.e, we have moved each coordinate by 0.5, is the same as a torus centered on $(0.5, 0.5)$, from symmetry. So the regions with same BPVs have the same probability, i.e $c_1 = c_4$ and $c_2 = c_3$, where c_i is the constant factor from the assumption. In addition, a torus that is rotated by 90 degrees is identical to the initial torus, which show $c_2 = c_1$. Therefore, all the regions have the same probability. ∎

Therefore, all we have to do is prove that $f(x)$ is not totally uniform. We proceed with proof by contradiction. Let us assume that the probability density function is constant over regions. By Lemma 4, we have that $f(x)$ is uniform over the entire torus. From the definition of the stationary probability, if we start a process $X(t)$ such that $X(0)$ is distributed with the stationary PDF $f(x)$, then for every $t > 0$ we have $X(t)$ distributed with the PDF $f(x)$. We show that for some small enough $t > 0$, $X(t)$ is not distributed uniformly, a contradiction. We start by dividing the U_1 square into 9 equal, square parts (See Figure 11). We denote the square in the i-th row, j-th column G_{ij}. We define a process $X(t)$, that acts according to $G(v)$, such that $X(0)$ is distributed uniformly. We prove that assuming $X(0)$ is uniform, $\Pr(X(t) \in G_{22}) < \Pr(X(t) \in G_{23})$ for some small enough $t > 0$, which disproves the assumption that $X(t)$ is uniform. We note, if it was uniform the probabilities would be $\Pr(X(t) \in G_{22}) = \Pr(X(t) \in G_{23}) = \mu(G_{22}) = \frac{1}{36}$. First, we prove an upper bound on $\Pr(X(t) \in G_{22})$, and then we prove a lower bound on $\Pr(X(t) \in G_{23})$.

Lemma 8 *The upper bound is, for a set G_{23}^L we define soon:*

$$\Pr(X(t) \in G_{22}) \ < \ \Phi(-c) + \Pr(X(t) \in G_{23} \mid X(0) \in G_{23}^L) \cdot \mu(G_{23}^L) + (\frac{1}{6} + 2vc\sqrt{t})(2c\sqrt{t}). \quad (3)$$

Where c is some arbitrary positive constant, v is the parameter from $G(v)$, and t is the time parameter.

First, we note that if we start at a point close to G_{22}, we have a high chance to get into G_{22} in time t. We define that set of points as follows:

$$G_{22}^* := \left[\frac{1}{6} - c \cdot \sqrt{t}, \frac{1}{3} + c \cdot \sqrt{t}\right] \times \left[\frac{1}{6} - v \cdot c \cdot \sqrt{t}, \frac{1}{3} + v \cdot c \cdot \sqrt{t}\right]$$

Where c is some arbitrary positive constant, v is the parameter from $G(v)$, and t is the time parameter.

The probability of entering if $X(0) \notin G_{22}^*$ is bounded by the following lemma.

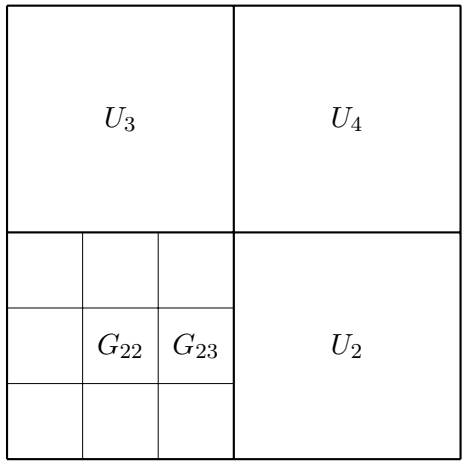

Figure 11: G_{22}, G_{23} in the torus

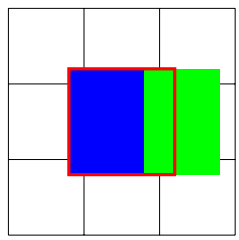

Figure 12: Blue is G_{22}^L, Red is G_{22}^*, Green is G_{23}^L

Lemma 9 $\Pr(X(t) \in G_{22} \mid X(0) \notin G_{22}^*) < \Phi(-c)$ - *i.e, the probability of finishing within G_{22} is very small, if we start far away from G_{22}.*

Proof: Let $X(0) \in \mathcal{S}$. We first prove the lemma with the assumption $\forall i \leq t, X(i) \in U_1$. We first assume $X(0)_2$ is closest to G_{22}^*, according to the definition of distance. Without loss of generality, $X(0)_2 > \frac{1}{3} + v \cdot c \cdot \sqrt{t}$. After shuffling some terms around, we have $\frac{\frac{1}{3} - X(0)_2}{v\sqrt{t}} < -c$. The projection of the process to the second coordinate is simply a Brownian motion with parameter v. It is known that Brownian motion is distributed as $\mathcal{N}(X(0), v\sqrt{t})$ after time t, and therefore $\Pr\left(X(t)_2 < \frac{1}{3}\right) = \Phi(\frac{\frac{1}{3} - X(0)_2}{v \cdot \sqrt{t}}) < \Phi(-c)$, where the last inequality is from a CDF being monotone. The proof is similar when $X(0)_1$ is closest, and we get the bound for the lemma. We note that to be in G_{22}, the $X(t)$ must be in the right interval in both axes, and thus one axis is enough to bound it.

For the case that $X(i) \notin U_1$ for some $i \leq t$, we note that

$$\Pr\left(X(t) \in G_{22} \mid X(i) \notin U_1\right) \leq \Pr(X(t) \in G_{22} \mid X(j) \in U_1 \setminus G_{22}^*, t > j > i) \ ,$$

from continuity - Since Brownian motion is continuous, it must pass through a point at which its at $U_1 \setminus G_{22}^*$ since G_{22}^* is in the interior of U_1. At that point, the proof from above applies. ∎

This lemma shows us we can simply bound the probability of reaching G_{22}, starting from anywhere in $\mathcal{T} \setminus G_{22}^*$, by the error function. Therefore we only have to bound $\Pr(X(t) \in G_{22} \mid X(0) \in G_{22}^*)$. To cut down on the area even further, an additional lemma can be used to help narrow down a bound

for a new area G_{22}^L. Denote by G_{22}^L the area defined by $G_{22}^L := G_{22}^* \cap \{(x,y) \mid x < \frac{1}{3} - c\sqrt{t}\}$.

Lemma 10 *For every $x = (x_1, x_2) \in G_{22}^L$,there is an equivalent $z := (x_1 + \frac{1}{6}, x_2)$ such that*

$$|\Pr(X(t) \in G_{22} \mid X(0) = x) - \Pr(X(t) \in G_{23} \mid X(0) = z)| < 2 \cdot \Phi(-c).$$

This lemma means that for a large chunk of G_{22}^*, we could bound the probability by an equivalent probability of reaching G_{23}. Since we are proving an upper bound, a direct comparison would help us narrow down the area that is different between the two groups.

Proof: We define a new process Y that acts like Brownian motion with parameters specified by U_1. We note the two processes are the same as long the motion does not reach the borders, and therefore $\Pr(\exists i \leq t.X(i) \notin U_1) = \Pr(\exists i \leq t.Y(i) \notin U_1)$. Let us look at $Y(0) = X(0)$. We look at $max(Y_i, t)$ and $min(Y_i, t)$ for $t = 1, 2$. To cross the boundaries of the grid into, U_2, we need to have $max(Y_1, t) > \frac{1}{2}$, or $min(Y_1, t) < 0$. By the reflection principle of Brownian motion, we have that $\Pr\left(max(Y_1, t) > \frac{1}{2}\right) = 2\Pr\left(Y(t)_1 \geq \frac{1}{2}\right)$. $Y(t)_1 \sim N\left(Y(0)_1, t\right)$ and therefore we get that $\Pr\left(max(Y_1, t) > \frac{1}{2}\right) \leq 2\Phi(-c)$. We get similar results for all other borders as well. ∎

We define the shifted area from above as $G_{23}^L := \{(x,y) \mid (x - \frac{1}{6}, y) \in G_{22}^L\}$. We generally use G_{23}^L henceforth.

Finally, for $G_{22}^* \setminus G_{22}^L$, we bound from above by 1. We note that the area of that set equals:

$$\begin{aligned}\mu\left(G_{22}^* \setminus G_{22}^L\right) &= \mu\left([\frac{1}{3} - c \cdot \sqrt{t}, \frac{1}{3} + c \cdot \sqrt{t}] \times [\frac{1}{6} - v \cdot c \cdot \sqrt{t}, \frac{1}{3} + v \cdot c \cdot \sqrt{t}]\right) = \\ &= (2vc\sqrt{t})(2c\sqrt{t})\end{aligned}$$

We get that, the probability is bounded by (3), showing the upper bound:

$$\begin{aligned}\Pr(X(t) \in G_{22}) = &(1 - \mu(G_{22}^*) \Pr\left[X(t) \in G_{22} \mid X(0) \notin G_{22}^*\right)] \\ &+ \mu(G_{22}^L) \cdot \Pr\left[X(t) \in G_{22} \mid X(0) \in G_{22}^L\right] \\ &+ \mu(G_{22}^* \setminus G_{22}^L) \Pr\left[X(t) \in G_{22} \mid X(0) \in \mu(G_{22}^* \setminus G_{22}^L)\right] \\ &< 8\Phi(-c) + \Pr\left[X(t) \in G_{23} \mid X(0) \in G_{23}^L\right] \cdot \mu(G_{23}^L) \\ &+ \left(\frac{1}{6} + 2vc\sqrt{t}\right)\left(2c\sqrt{t}\right).\end{aligned}$$

We now find a lower bound for $\Pr\left(X(t) \in G_{23}\right)$. Again, we seperate into cases for each $X(0)$. For $X(0) \in G_{23}^L$, we simply can use the lemma Lemma 10 from before. For other $X(0)$ we need to bound some other way. The bound is:

Lemma 11 *There is a lower bound for* $\Pr\left(X(t) \in G_{23}\right)$.

$$\begin{aligned}\Pr\left(X(t) \in G_{23}\right) > &\Pr(X(t) \in G_{23} \mid X(0) \in G_{23}^L) \cdot \mu(G_{23}^L) - \Phi(-c) \\ &+ \frac{2 \cdot \Phi(-d)}{3} \cdot \left(\frac{1}{6} - 2vc\sqrt{t}\right)\left(vd\sqrt{t}\right).\end{aligned} \tag{4}$$

The terms at the end come from a new area we prove the lower bound of, H_{23}. We define H_{23} as follows:

$$H_{23} = [\frac{1}{2}, \frac{1}{2} + v \cdot d \cdot \sqrt{t}] \times [\frac{1}{6} + v \cdot c \cdot \sqrt{t}, \frac{1}{3} - v \cdot c \cdot \sqrt{t}].$$

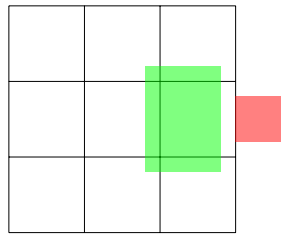

Figure 13: Red is H_{23}, Green is G^L_{23}

We note that this area is disjoint to G^L_{23}, and thus when calculating the total probability, it will not be counted twice.

Lemma 12 *There is a lower bound for* $\Pr\left(X(t) \in G_{23} \mid X(0) \in H_{23}\right)$,

$$\Pr(X(t) \in G_{23} \mid X(0) \in H_{23}) > (1 - \Phi(-c))\,(\frac{2\Phi(-d)}{3}).$$

Proof: We prove the lower bound by treating $X_1(t)$ and $X_2(t)$ independently. For X_2, by using the same argument of the Gaussian distribution we used before, we get $\Pr\left(X_2(t) \notin \left[\frac{1}{6}, \frac{1}{3}\right]\right) \leq \Phi(-c)$. We took the additional v in the definition of H_{23} to make sure its true for all regions - Both with parameter 1 and parameter v.

For X_1, we prove it a different way. If $X(t) \in G_{23}$, than it must mean $\frac{1}{3} \leq X_1(t) \leq \frac{1}{2}$. From continuity of Brownian motion, and from definition of the space, than if we started in U_2 it must either mean $X(i) = 0$ or $X(i) = \frac{1}{2}$, for some $i \leq t$. Let us assume only the second case - I.e, we bound the first case by 0.

$$\Pr\left(\frac{1}{3} \leq X(t)_1 < \frac{1}{2} \mid \frac{1}{2} \leq X(0)_1 < \frac{1}{2} + d \cdot v \cdot \sqrt{t}\right) \geq \Pr\left(\frac{1}{3} \leq X(t)_1 < \frac{1}{2} \mid i \leq t, X(i)_1 = \frac{1}{2}\right) \cdot \Pr\left(i \leq t, X(i)_1 = \frac{1}{2} \mid \frac{1}{2} \leq X(0)_1 < \frac{1}{2} + d \cdot v \cdot \sqrt{t}\right).$$

We bound the two probabilities separately. For $\Pr\left(i \leq t, X(i)_1 = \frac{1}{2} \mid \frac{1}{2} \leq X(0)_1 < \frac{1}{2} + d \cdot v \cdot \sqrt{t}\right)$, we can use the same fact from Lemma 10 we did - We define a new process Y to be a standard Brownian process with the parameters defined by U_2. We want to find $\Pr\left(i \leq t, X(i)_1 = \frac{1}{2} \mid \frac{1}{2} \leq X(0)_1 < \frac{1}{2} + d \cdot v \cdot \sqrt{t}\right)$. But this is simply $\Pr\left(Min(X_1, t) \leq \frac{1}{2}\right)$. As before, this equals $\Pr\left(Min(Y_1, t) \leq \frac{1}{2}\right)$. This is distributed normally upon defining $Y(0)_1$, so we bound it by $Y(0)_1 \in H_{23}$ such that the Gaussian has the hightest mean, for $Y(0)_1 = \frac{1}{2} + d \cdot v \cdot \sqrt{t}$. The standard deviation is always $v\sqrt{t}$, which nets us a probability of $2\Phi(-d)$.

For the other probability, $\Pr(\frac{1}{3} < X(t)_1 < \frac{1}{2} \mid X(i)_1 = \frac{1}{2})$. We first note this can be written as a conditional probability, when we condition it on i, as follows -

$$\begin{aligned}\Pr(\frac{1}{3} < X(t)_1 < \frac{1}{2} \mid X(i)_1 = \frac{1}{2}) &= \int_0^t \Pr(\frac{1}{3} < X(t)_1 < \frac{1}{2} \mid X(i)_1 = \frac{1}{2}) \Pr(i = a) da \\ &= \int_0^t \Pr(\frac{1}{3} < X(t-i)_1 < \frac{1}{2} \mid X(0)_1 = \frac{1}{2}) \Pr(i = a) da\end{aligned}$$

. By the definition of the border regions of the torus, when $t \to 0$ the probability of ending up in the left side of the border goes to $\frac{1}{2}$. Namely, for small enough t we have $\Pr\left(\frac{1}{3} \leq X(t-i)_1 < \frac{1}{2} \mid, X(0)_1 = \frac{1}{2}\right) > \frac{1}{3}$. Therefore we can bound the integral by $\int_0^t \frac{\Pr(i=a)}{3} da = \frac{1}{3}$. We note that for small t the probability of reaching $\frac{1}{3}$ is exponentially small based on t and thus can be ignored. So we can write that as

$$\Pr\left(X(t)_1 < \frac{1}{2} \mid X(0)_1 \leq \frac{1}{2} + d \cdot v \cdot \sqrt{t}\right) \geq \frac{1}{3} \cdot 2 \cdot \Phi(-d)$$

Which is the bound we wanted to show. Therefore, we bound with

$$\begin{aligned}\Pr\left(X(t) \in G_{23} \mid X(0) \in H_{23}\right) = &\Pr\left(\frac{1}{3} \leq X(t)_1 < \frac{1}{2} \mid \frac{1}{2} \leq X(0)_1 < \frac{1}{2} + d \cdot v \cdot \sqrt{t}\right) \\ &\cdot \Pr\left(\frac{1}{3} \leq X(t)_2 < \frac{1}{2} \mid \frac{1}{6} + v \cdot c \cdot \sqrt{t} \leq X(0)_2 \leq \frac{1}{3} - v \cdot c \cdot \sqrt{t}\right)\end{aligned}$$

Which, as we have proven equals to:

$$(1 - \Phi) \cdot \frac{2\Phi(-d)}{3}$$

∎

We bound the rest of the torus by 0 - $\Pr(X(t) \in G_{23} \mid X(0) \notin (H_{23} \cup G_{23}^L) \geq 0$. Therefore, we are bounded from below by (4).

Therefore to prove the full theorem, we compare the bounds we got, (4) and (3). We get:

$$(\frac{1}{6} + 2vc\sqrt{t})(2c\sqrt{t}) + 2\Phi(-c) < \frac{2 \cdot \Phi(-d)}{3} \cdot (\frac{1}{6} - 2vc\sqrt{t})(vd\sqrt{t})$$

For small enough d, large enough c, small enough t and large enough v such that this is more than the area from before, and we are done. Therefore we have a contradiction to the PDF of the torus being uniform over every region in the case of a grid with BPVs $(1, v)$ on one diagonal and $(v, 1)$ on the other.